

AeroRAG

A retrieval-augmented co-pilot for FAA regulations, NTSB incidents, and live aviation weather.

Plan window: Fri May 22 – Tue Jun 30, 2026 · 6 weeks · 28 weekdays

Cadence: Mon–Fri only

Repository: <https://github.com/v1gneshn/AeroRAG>

Generated: 2026-05-21

*One scoped weekly outcome. One commit per workday.
Weekends off.*

Project summary

AeroRAG is a six-week prototype that builds a retrieval-augmented Q&A system over three layers of aviation data — federal regulations, historical incident reports, and live weather — with strict citation enforcement and an honest, measured eval loop. The plan ships a deployable **v0.1.0** by June 30. Work happens Monday through Friday; weekends are explicitly off.

Phases

Phase	Window	What good looks like
Phase 1 — Reg-RAG	Weeks 1–3 (May 22 – Jun 12)	Trustworthy FAR / AIM Q&A with strict citations.
Phase 2 — Incident-RAG	Week 4 (Jun 15 – Jun 19)	NTSB pattern search with metadata filters.
Phase 3 — Wx + Ship	Weeks 5–6 (Jun 22 – Jun 30)	METAR/TAF briefings + deployed, eval'd, released.

Stack

Language	Python 3.11
Orchestration	LangChain (kept minimal)
Vector store	ChromaDB (file-backed)
Sparse retrieval	rank_bm25 + RRF fusion
Re-ranker	sentence-transformers cross-encoder
Generator	OpenAI gpt-4o-mini
Embeddings	text-embedding-3-small (1536-d)
UI	Streamlit
Eval	RAGAS + LLM-judge + golden set
Deploy	Docker → HF Spaces / Fly.io

Weekly milestones

Date	Milestone
Fri May 29	Corpus parsed & manifested
Fri Jun 05	First retrieval baseline measured
Fri Jun 12	Cited answers, hybrid retrieval, faithfulness eval
Fri Jun 19	Streamlit UI v1 + NTSB ingested
Fri Jun 26	Unified router, conversational, RAGAS sweep
Tue Jun 30	v0.1.0 deployed, demo video, post-mortem

Weekly plan

Each week ends with a single, demoable outcome. Anything that doesn't fit the week gets cut, not pushed forward.

Week 1

Foundations & corpus

Fri May 22 + Mon–Fri May 25–29 · 6 weekdays

- Bootstrap venv; verify OpenAI access with gpt-4o-mini
- Source-map FAR (14 CFR) / AIM / PHAK; download into data/raw/ with SHA256 manifest
- PDF parsing (pdfplumber) + ecFR XML parser that preserves §/(a)/(1) hierarchy
- Spot-check 5 parsed sections vs. source PDFs to catch parser gotchas

***End-of-week:** Authoritative corpus parsed into structured JSONL with verifiable provenance.*

Week 2

Chunking, embeddings, first retrieval

Mon–Fri Jun 01–05 · 5 weekdays

- Two chunkers: recursive (800/120) + regulatory §/(a)/(1) boundary
- Embed with text-embedding-3-small → persist Chroma collection
- CLI: aerorag search " returns top-5 with citations
- Hand-craft a 10-question golden set; baseline recall@5 for both chunkers

***End-of-week:** Working Chroma index + 10-question golden set with baseline numbers.*

Week 3

RAG pipeline + hybrid retrieval

Mon–Fri Jun 08–12 · 5 weekdays

- Strict citation prompt; first end-to-end answer with §-level cites
- Citation enforcement: reject + regenerate (max 2 attempts) or refuse
- BM25 sparse index + RRF hybrid fusion with dense retrieval
- Cross-encoder re-rank top-25 → top-5
- LLM-judge faithfulness eval on golden set

***End-of-week:** Cited reg Q&A; beating dense baseline by ≥5pp recall@5; faithfulness measured.*

Week 4

Streamlit UI + NTSB ingestion

Mon–Fri Jun 15–19 · 5 weekdays

- Streamlit UI v1: question box, answer, expandable citations panel
- Source PDF page links from each citation
- Pull NTSB CAROL dataset (last 10 yrs) + incident schema
- Embed narratives; structured fields as metadata filters
- Metadata-filtered retrieval (aircraft model, flight phase, severity)

***End-of-week:** Clickable demo for regs + a filterable NTSB index seeded.*

Week 5**Router + memory +
METAR/TAF + full eval**

Mon–Fri Jun 22–26 · 5 weekdays

- Query router: classify reg / incident / wx
- Pattern-finding answer for incident clusters
- Conversational memory (last 4 turns)
- METAR/TAF decoder tool (aviationweather.gov) + Wx briefing chain w/ non-dispatch disclaimer
- RAGAS full sweep across all three sub-domains

End-of-week: Unified UX that routes across three data sources; system-wide eval table.

Week 6**Polish & ship**

Mon–Tue Jun 29–30 · 2 weekdays

- Latency profiling + LLM-response disk cache; Dockerize
- Deploy to HF Spaces or Fly.io; 3-min demo screencast; tag v0.1.0 with post-mortem

End-of-week: Public URL, demo video, GitHub Release with honest post-mortem.

Working rhythm

Mon–Fri. Weekends off. ~1.5 to 2 hrs per workday. One commit per day with a short message and a one-paragraph daily log. Even a "nothing worked today, here's what I learned" log counts — cadence beats heroics.

```
./scripts/daily.sh "Wk 2 / Day 3 · embeddings + Chroma persist"
```

→ stamps today's log from the template, opens it, stages, commits.

Anti-goals

Things this plan refuses to do, so the scope stays honest:

- **No fine-tuning.** Retrieval-only — that's where the leverage is for v0.1.
- **No agent frameworks that hide what's happening.** A single classifier-router is the limit.
- **No custom chat UI.** Streamlit until it can't keep up.
- **No mid-week scope expansion.** Ideas go to docs/parking_lot.md, not into this week.
- **No silent failures.** Every answer is either cited or refused.

After v0.1.0 (stretch / v0.2 candidates)

- Multi-hop retrieval (sub-question decomposition).
- Re-rank ablation: BGE vs. cross-encoder vs. ColBERT.
- Swap to local embeddings (bge-small-en-v1.5) to drop OpenAI dependency.
- Whisper voice input for hands-free cockpit-style queries.
- Publish eval harness as a Hugging Face dataset for reproducibility.